

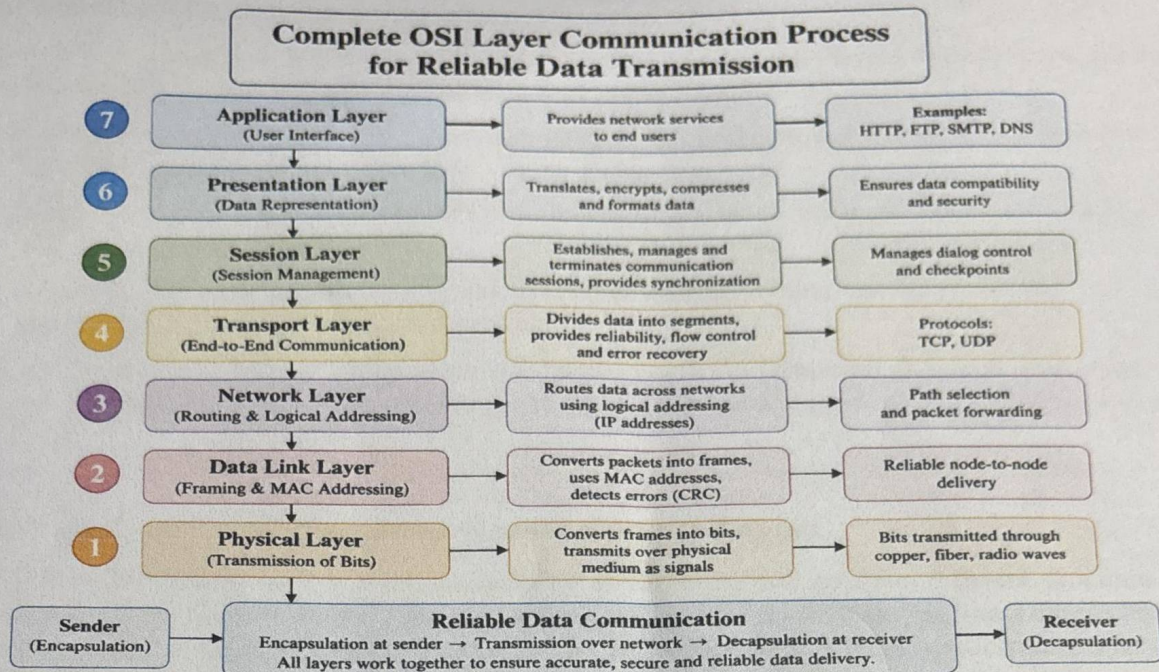
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COURSE NAME : COMPUTER NETWORKS

Concept Mapping and Analysis of OSI Layer Communication in Computer Networks

Q1. Complete OSI Layer Communication Process for Reliable Data Transmission

The Open Systems Interconnection (OSI) model is a standardized seven-layer architecture that enables reliable communication between devices connected over a network. Each layer performs a specific function while interacting with the adjacent layers to ensure that data is transmitted accurately, securely, and efficiently. In the given concept map, the missing layers are the Session Layer and the Data Link Layer, completing the sequence: Application → Presentation → Session → Transport → Network → Data Link → Physical. The Application Layer provides network services to end users through protocols such as HTTP, FTP, and SMTP. The Presentation Layer translates, compresses, and encrypts data to maintain compatibility and security between communicating systems. The Session Layer establishes, manages, and terminates communication sessions while providing synchronization points that allow interrupted transmissions to resume without restarting the entire process. The Transport Layer divides data into segments, performs error recovery, flow control, and reliable end-to-end communication using protocols such as TCP. The Network Layer determines the best path for data transmission through logical addressing and routing using IP addresses. The Data Link Layer converts packets into frames, performs physical addressing using MAC addresses, detects transmission errors through mechanisms such as CRC, and ensures reliable node-to-node communication. Finally, the Physical Layer converts digital data into electrical, optical, or radio signals for transmission over the physical medium. During communication, data undergoes encapsulation at the sender, where each layer adds its own control information, and decapsulation at the receiver, where these headers are removed in reverse order to recover the original message. The coordinated operation of all seven layers ensures interoperability between different network devices, minimizes transmission errors, enhances security, and guarantees reliable delivery of information across modern computer networks.



Concept Map Architecture (Q1)

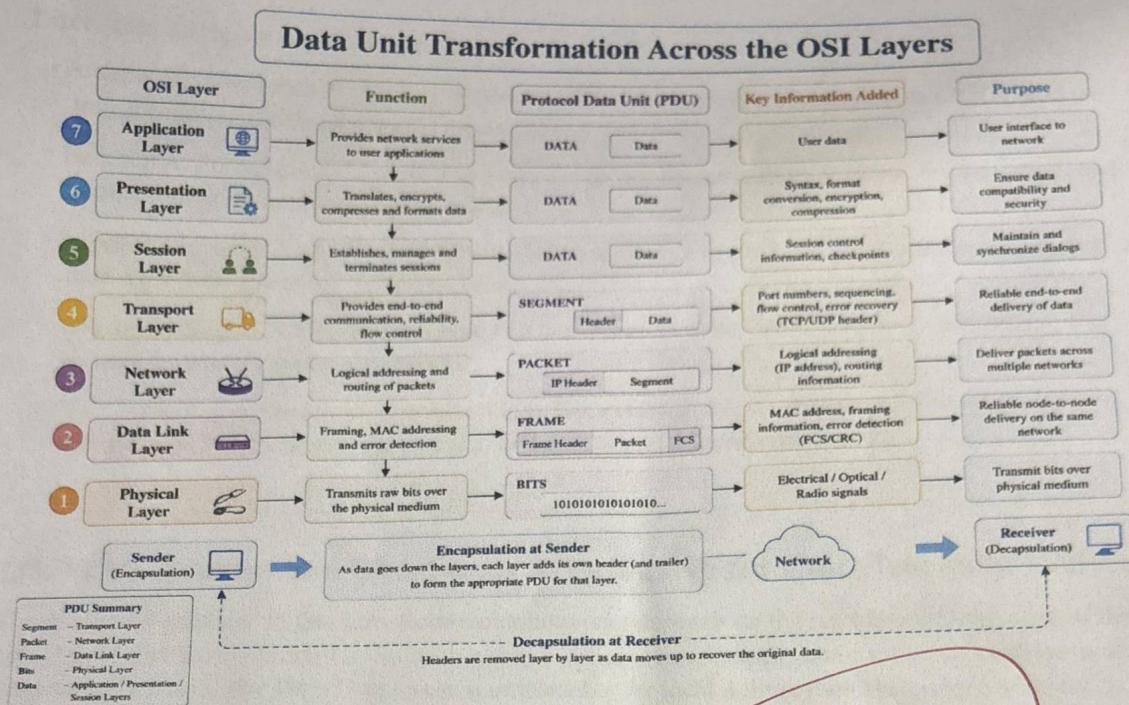
Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
Application Layer	sends user requests to	Presentation Layer
Presentation Layer	translates and encrypts data for	Session Layer
Session Layer	establishes and manages	Communication Session
Communication Session	transfers data through	Transport Layer
Transport Layer	divides data into	Segments
Transport Layer	provides	Flow Control and Error Recovery
Network Layer	routes data using	IP Address
Data Link Layer	converts packets into	Frames
Data Link Layer	identifies devices using	MAC Address
Physical Layer	transmits	Bits over Transmission Media
All OSI Layers	perform	Reliable Data Communication

Literature Insights & Evidence (Q1)

- Andrew S. Tanenbaum & David J. Wetherall (2021): Explain the seven-layer OSI architecture and the functions performed by each layer during network communication.
 - Behrouz A. Forouzan (2022): Describes encapsulation, decapsulation, and the responsibilities of every OSI layer in reliable data transmission.
 - James F. Kurose & Keith W. Ross (2021): Discuss the interaction between protocol layers and their role in end-to-end communication.
 - William Stallings (2020): Explains logical addressing, framing, routing, and error detection mechanisms in layered network architecture.
 - Cisco Networking Academy – Introduction to Networks (2023): Demonstrates how the OSI model supports interoperability, troubleshooting, and reliable communication in modern computer networks.
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Q2. Data Unit Transformation Across the OSI Layers

The OSI model enables reliable communication by transforming user data into different data units as it moves through the network layers. Each layer performs a specific function and assigns a unique Protocol Data Unit (PDU) that supports efficient transmission. The process begins at the Application, Presentation, and Session Layers, where the information is treated simply as Data because these layers focus on user services, data formatting, encryption, compression, and session management. When the information reaches the Transport Layer, it is divided into Segments, allowing reliable end-to-end communication through sequencing, flow control, and error recovery. The Network Layer encapsulates each segment into a Packet by adding logical addressing information such as IP addresses, enabling routers to determine the best path across interconnected networks. The Data Link Layer further encapsulates packets into Frames by adding MAC addresses and error detection information such as Frame Check Sequence (FCS), ensuring reliable node-to-node communication. Finally, the Physical Layer converts frames into Bits, represented as electrical, optical, or radio signals that travel through the transmission medium. At the receiving device, the reverse process known as decapsulation removes the headers added by each layer and reconstructs the original user data. This systematic transformation of data units provides modularity, error control, logical addressing, physical addressing, and reliable communication across heterogeneous network environments. The encapsulation and decapsulation process allows devices using different hardware and operating systems to communicate seamlessly while maintaining data integrity, efficiency, and interoperability.



Concept Map Architecture (Q2)

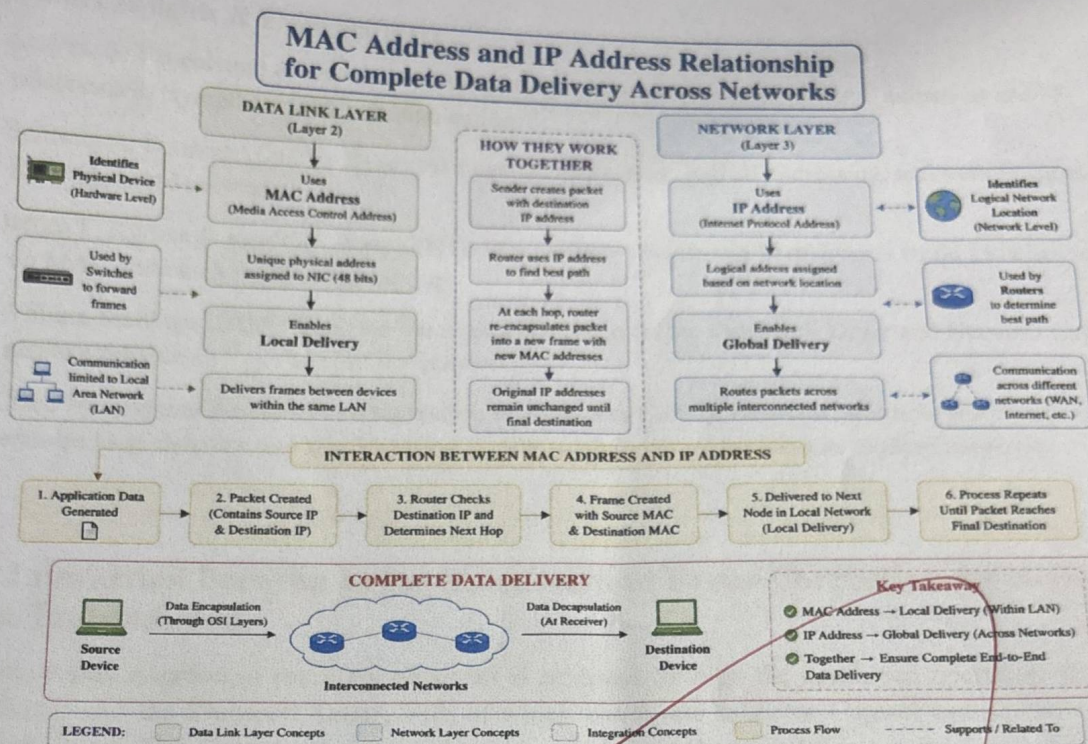
Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
Application Layer	generates	Data
Presentation Layer	formats and encrypts	Data
Session Layer	manages	Communication Session
Transport Layer	divides data into	Segments
Segments	are encapsulated into	Packets
Network Layer	adds	IP Address
Packets	are converted into	Frames
Data Link Layer	adds	MAC Address and FCS
Frames	are converted into	Bits
Physical Layer	transmits	Bits through Transmission Media
Receiver	performs	Decapsulation to recover Data

Literature Insights & Evidence (Q2)

- Andrew S. Tanenbaum & David J. Wetherall (2021): Explain protocol data units (PDUs) and the transformation of data across OSI layers.
 - Behrouz A. Forouzan (2022): Describes encapsulation, decapsulation, framing, segmentation, and packet formation in layered communication.
 - James F. Kurose & Keith W. Ross (2021): Discuss the role of protocol layers and their corresponding data units during network communication.
 - William Stallings (2020): Explains how PDUs support modular communication, addressing, and reliable transmission across networks.
 - Cisco Networking Academy – Introduction to Networks (2023): Demonstrates the relationship between OSI layers, protocol data units, and end-to-end data delivery.
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Q3. MAC Address and IP Address Relationship for Complete Data Delivery

The successful delivery of data across computer networks depends on the coordinated operation of the Data Link Layer and the Network Layer, each using a different addressing mechanism to perform a specific communication task. The Data Link Layer is responsible for local delivery within a single network or Local Area Network (LAN). It uses MAC (Media Access Control) addresses, which are unique physical addresses permanently assigned to network interface cards (NICs). MAC addresses ensure that frames are delivered correctly between devices connected to the same physical network. However, MAC addresses alone cannot communicate beyond the local network. To enable communication between different networks, the Network Layer uses IP (Internet Protocol) addresses, which are logical addresses assigned to devices based on their network location. Routers examine IP addresses to determine the optimal path for forwarding packets from the source network to the destination network. During transmission, the sender first determines the destination IP address and identifies the appropriate route. When a packet reaches each intermediate network, the router encapsulates it into a new frame with updated source and destination MAC addresses while keeping the original IP addresses unchanged until the packet reaches its final destination. This interaction between IP addressing for global routing and MAC addressing for local delivery ensures efficient communication across interconnected networks. The combination of logical and physical addressing enables scalable, reliable, and accurate data transmission while supporting internetworking, routing, and interoperability among heterogeneous network devices. Together, the Data Link Layer and Network Layer provide the foundation for complete end-to-end communication in modern computer networks.



Concept Map Architecture (Q3)

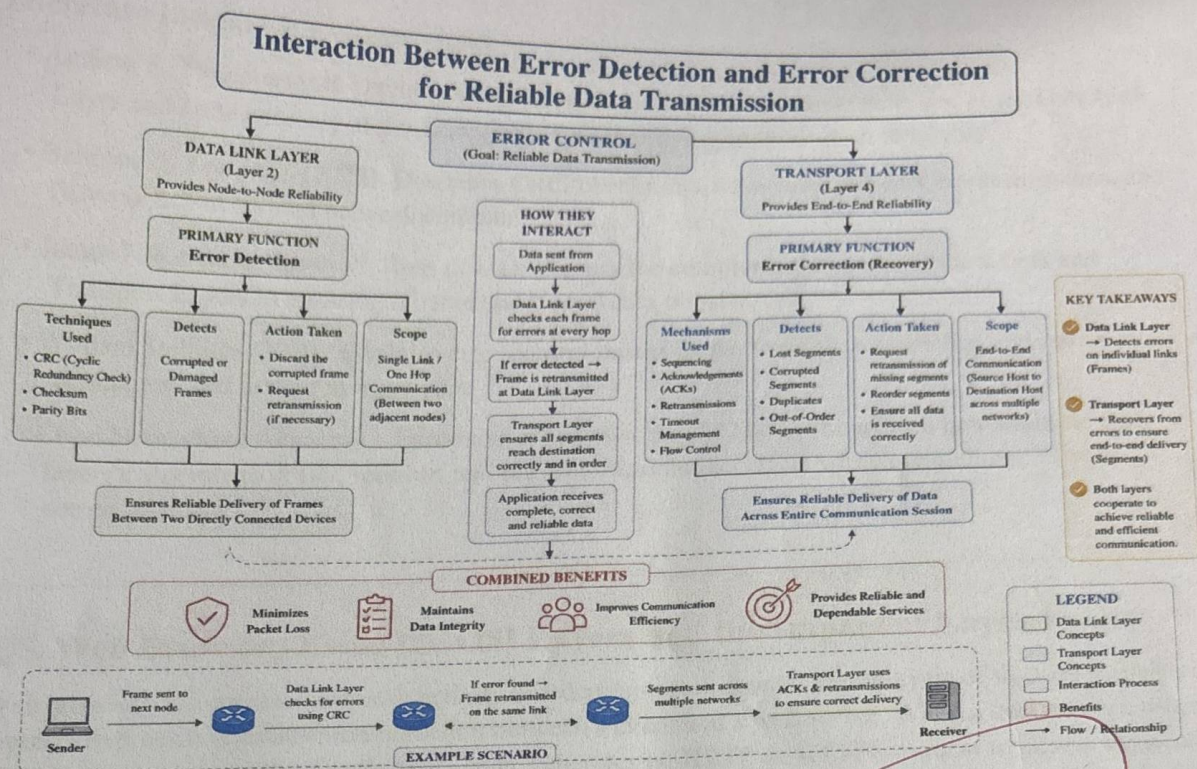
Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
Data Link Layer	uses	MAC Address
MAC Address	enables	Local Delivery
Local Delivery	forwards frames within	Local Area Network (LAN)
Network Layer	uses	IP Address
IP Address	enables	Global Delivery
Global Delivery	routes packets across	Multiple Networks
Router	examines	IP Address
Router	updates	MAC Address for Each Hop
MAC Address	identifies	Physical Device
IP Address	identifies	Logical Network Location
MAC Address and IP Address	work together to ensure	Complete Data Delivery

Literature Insights & Evidence (Q3)

- Andrew S. Tanenbaum & David J. Wetherall (2021): Explain the roles of MAC addresses and IP addresses in layered network communication.
 - Behrouz A. Forouzan (2022): Describes physical addressing, logical addressing, and routing across interconnected networks.
 - James F. Kurose & Keith W. Ross (2021): Discuss how routers use IP addresses while switches rely on MAC addresses for data forwarding.
 - William Stallings (2020): Explains the interaction between the Data Link Layer and Network Layer to achieve reliable end-to-end communication.
 - Cisco Networking Academy – Introduction to Networks (2023): Demonstrates how MAC addressing enables local delivery and IP addressing enables global communication in modern networks.
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Q4. Interaction Between Error Detection and Error Correction for Reliable Data Transmission

Reliable communication in computer networks is achieved through the combined operation of the Data Link Layer and the Transport Layer, both of which contribute to Error Control but perform different responsibilities. The Data Link Layer provides reliable node-to-node communication by detecting transmission errors that occur on a single physical link. It accomplishes this using techniques such as Cyclic Redundancy Check (CRC), checksums, and parity bits. When a damaged frame is detected, the receiver discards the frame and requests retransmission if necessary. However, the Data Link Layer only ensures reliability between two directly connected devices and does not guarantee successful delivery across multiple networks. The Transport Layer extends reliability by providing end-to-end communication between the source and destination hosts. Protocols such as Transmission Control Protocol (TCP) perform sequencing, acknowledgements (ACKs), retransmissions, timeout management, and flow control to recover lost, delayed, or corrupted segments. While the Data Link Layer identifies errors within individual frames, the Transport Layer ensures that every segment reaches the destination correctly and in the proper order. These two layers work together to minimize packet loss, maintain data integrity, improve communication efficiency, and provide dependable network services. During transmission, the Data Link Layer protects each hop of the journey, whereas the Transport Layer supervises the complete communication session from sender to receiver. Their coordinated interaction forms the foundation of reliable data transmission and enables modern applications such as web browsing, file transfer, video conferencing, and online transactions to operate accurately and efficiently over computer networks.



Concept Map Architecture (Q4)

Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
Error Control	includes	Data Link Layer
Data Link Layer	performs	Error Detection
Error Detection	uses	CRC and Checksum
Error Detection	identifies	Corrupted Frames
Error Control	also includes	Transport Layer
Transport Layer	performs	Error Correction
Error Correction	uses	ACKs and Retransmissions
Transport Layer	provides	End-to-End Reliability
Data Link Layer	provides	Node-to-Node Reliability
Both Layers	work together for	Reliable Data Transmission
Reliable Data Transmission	ensures	Data Integrity and Successful Delivery

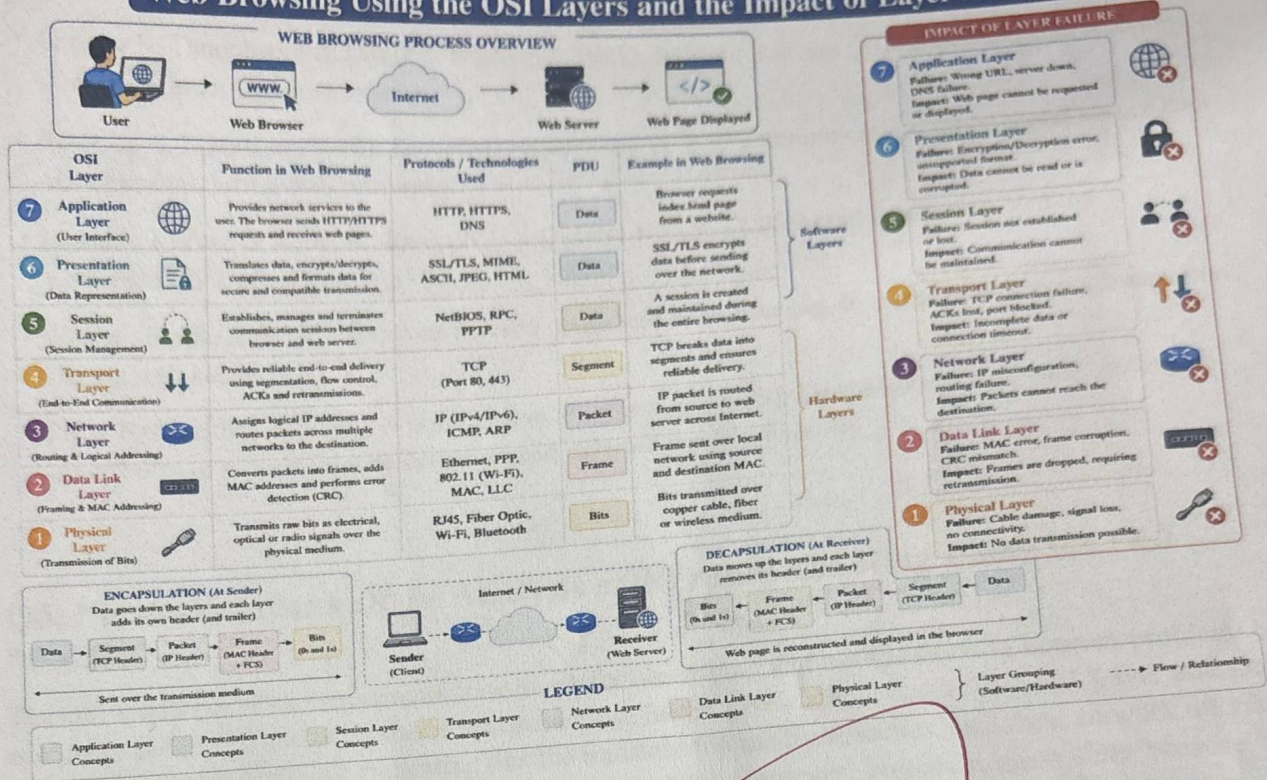
Literature Insights & Evidence (Q4)

- Andrew S. Tanenbaum & David J. Wetherall (2021): Explain how error detection at the Data Link Layer and error recovery at the Transport Layer improve communication reliability.
- Behrouz A. Forouzan (2022): Describes CRC, checksums, acknowledgements, retransmissions, and flow control in layered network communication.
- James F. Kurose & Keith W. Ross (2021): Discuss the complementary roles of Data Link and Transport Layers in ensuring reliable end-to-end data transfer.
- William Stallings (2020): Explains error control mechanisms, framing, sequencing, and retransmission techniques used in modern networks.
- Cisco Networking Academy – Introduction to Networks (2023): Demonstrates how multiple OSI layers cooperate to detect, recover, and prevent transmission errors while maintaining reliable communication.

Q5. Web Browsing Using the OSI Layers and the Impact of Layer Failure

Web browsing is a common network activity that demonstrates how all seven layers of the OSI model work together to provide reliable communication between a client and a web server. When a user enters a website address into a browser, the Application Layer generates an HTTP or HTTPS request to access the desired web resource. The Presentation Layer translates the data into a common format, compresses it if necessary, and encrypts it using SSL/TLS for secure communication. The Session Layer establishes, manages, and terminates the communication session between the client and server while maintaining synchronization throughout the exchange. The Transport Layer uses the Transmission Control Protocol (TCP) to divide data into segments, assign sequence numbers, provide acknowledgements, and perform retransmissions when packet loss occurs, ensuring reliable end-to-end communication. The Network Layer assigns logical IP addresses and determines the most efficient route through routers to reach the destination server. The Data Link Layer converts packets into frames, attaches source and destination MAC addresses, performs error detection using the Frame Check Sequence (FCS), and ensures reliable node-to-node communication. Finally, the Physical Layer transmits the data as electrical, optical, or wireless signals across the communication medium. At the receiving end, the reverse process of decapsulation reconstructs the original web page for display in the browser. If any OSI layer fails, communication is interrupted. For example, failure of the Physical Layer prevents signal transmission, a Data Link Layer failure causes frame errors, a Network Layer failure prevents routing, a Transport Layer failure causes unreliable communication, and an Application Layer failure makes web services inaccessible. Therefore, every OSI layer contributes to successful web browsing, making the layered architecture essential for reliable, secure, and efficient Internet communication.

Web Browsing Using the OSI Layers and the Impact of Layer Failure



Concept Map Architecture (Q5)

Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
User	requests	Web Page
Web Browser	communicates using	Application Layer (HTTP/HTTPS)
Presentation Layer	encrypts and formats	Web Data
Session Layer	establishes	Communication Session
Transport Layer	provides	Reliable TCP Communication
Network Layer	routes packets using	IP Address
Data Link Layer	delivers frames using	MAC Address
Physical Layer	transmits	Bits over Transmission Media
Receiver	performs	Decapsulation
Failure of Any Layer	interrupts	Communication Process
All OSI Layers	work together for	Successful Web Browsing

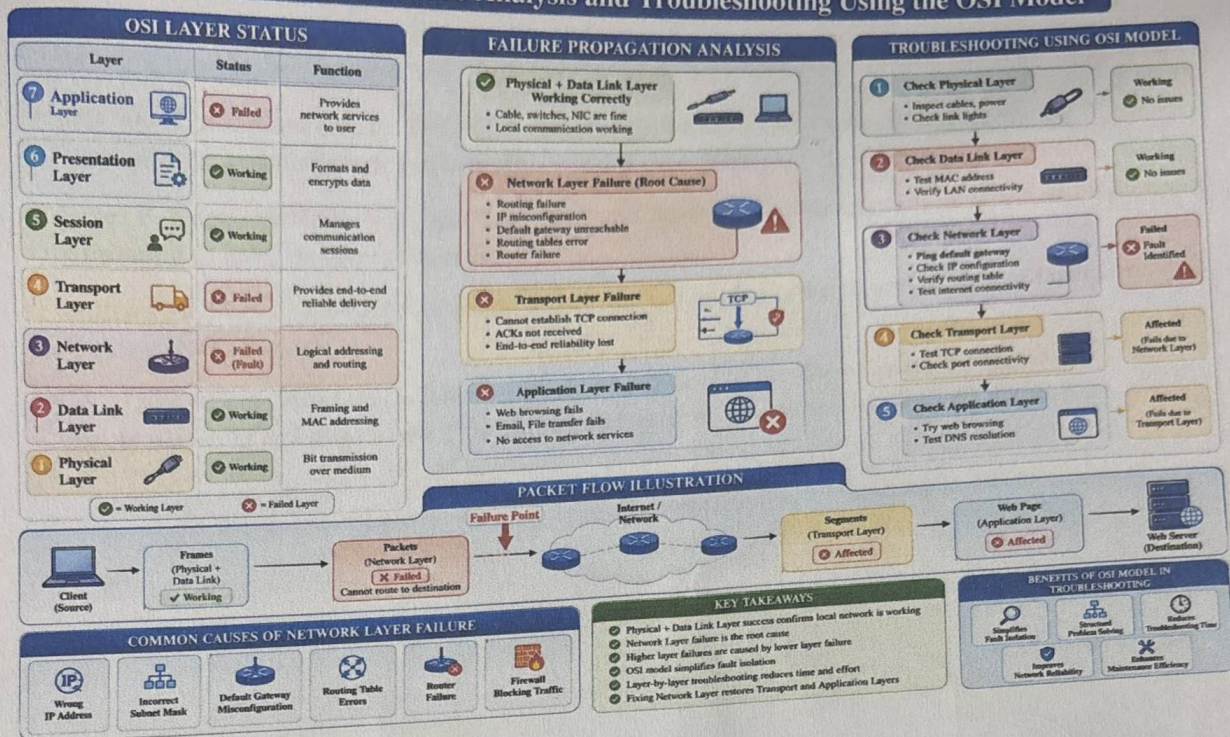
Literature Insights & Evidence (Q5)

- Andrew S. Tanenbaum & David J. Wetherall (2021): Explain how the OSI model supports Internet communication and web services.
- Behrouz A. Forouzan (2022): Describes encapsulation, TCP communication, routing, and secure data transmission during web browsing.
- James F. Kurose & Keith W. Ross (2021): Discuss HTTP, TCP/IP communication, and the responsibilities of each network layer.
- William Stallings (2020): Explains layered communication, routing, framing, and protocol interaction during web access.
- Cisco Networking Academy – Introduction to Networks (2023): Demonstrates how each OSI layer contributes to successful web browsing and network troubleshooting.

Q6. Network Layer Failure Analysis and Troubleshooting Using the OSI Model

The OSI model provides a systematic framework for identifying and troubleshooting network problems by analyzing each communication layer independently. In the given scenario, the Physical Layer and Data Link Layer operate correctly, indicating that the transmission medium, cables, switches, network interface cards, and local frame delivery are functioning properly. However, communication fails because the Network Layer is not operating correctly. The Network Layer is responsible for logical addressing, packet forwarding, and routing using Internet Protocol (IP). When this layer fails, routers cannot determine a valid route to the destination network, preventing packets from reaching their intended destination. Since the Transport Layer depends on successful packet delivery by the Network Layer, protocols such as TCP cannot establish reliable end-to-end communication, resulting in connection failures and the inability to exchange data. Consequently, the Application Layer also fails because applications such as web browsers, email clients, file transfer services, and cloud applications rely on the lower layers to deliver data successfully. By following the OSI troubleshooting approach, network administrators can isolate the fault efficiently instead of checking the entire network simultaneously. They first verify that physical connectivity and local communication are functioning correctly, then identify routing problems within the Network Layer by examining IP configuration, routing tables, gateways, or router functionality. Once the Network Layer issue is resolved, the Transport Layer automatically resumes reliable communication, allowing application services to function normally again. This layered troubleshooting methodology reduces diagnostic time, simplifies fault isolation, and improves the reliability and maintainability of modern computer networks.

Network Layer Failure Analysis and Troubleshooting Using the OSI Model



Concept Map Architecture (Q6)

Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
Physical Layer	successfully provides	Signal Transmission
Data Link Layer	successfully provides	Frame Delivery
Physical Layer and Data Link Layer	indicate	Local Network is Working
Network Layer	failure causes	Routing Failure
Routing Failure	prevents	Packet Delivery
Transport Layer	depends on	Network Layer
Transport Layer Failure	prevents	End-to-End Communication
Application Layer	depends on	Transport Layer
Application Layer Failure	prevents	User Access to Network Services
OSI Model	helps in	Layer-by-Layer Troubleshooting
Layer-by-Layer Troubleshooting	identifies	Network Layer as the Fault

Literature Insights & Evidence (Q6)

- Andrew S. Tanenbaum & David J. Wetherall (2021): Explain how the OSI model supports systematic network troubleshooting and fault isolation.
- Behrouz A. Forouzan (2022): Describes the responsibilities of the Network Layer, routing mechanisms, and diagnostic procedures.
- James F. Kurose & Keith W. Ross (2021): Discuss dependency among protocol layers and the impact of routing failures on communication.
- William Stallings (2020): Explains logical addressing, routing failures, and the use of layered models in network diagnostics.
- Cisco Networking Academy – Introduction to Networks (2023): Demonstrates practical troubleshooting techniques using the OSI model to identify and resolve network failures.

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